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Semester:

4th Semester

Subject:

Database Applications

Lab Task:

MySQL Lab Guide

Submitted To:

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Lab Task 5

Lab Tasks

1.Create the above tables with constraints.

The screenshot shows the phpMyAdmin web interface. The top navigation bar indicates the current context: Server: 127.0.0.1:3307, Database: college_db, and Table: employees. The left sidebar displays a tree view of the database structure, including 'college_db' and its tables 'courses' and 'employees'. The main area is the SQL query editor, which contains the following SQL statement:

```
1 CREATE TABLE departments (  
2   dept_id INT PRIMARY KEY,  
3   dept_name VARCHAR(100) UNIQUE  
4 );
```

Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. A checkbox for 'Bind parameters' is also present. At the bottom, there is a 'Delimiter' dropdown set to semicolon, and several checkboxes: 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked). A 'Go' button is located at the bottom right of the query editor area.

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0207 seconds.)

```
CREATE TABLE departments ( dept_id INT PRIMARY KEY, dept_name VARCHAR(100) UNIQUE );
```

[Edit inline] [Edit] [Create PHP code]

Console

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
- information_schema
- mysql
- performance_schema
- phpmyadmin

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.employees:

```
1 CREATE TABLE students (  
2     student_id INT AUTO_INCREMENT PRIMARY KEY,  
3     name VARCHAR(100) NOT NULL,  
4     email VARCHAR(100) UNIQUE,  
5     age INT,  
6     dept_id INT,  
7     FOREIGN KEY (dept_id) REFERENCES departments(dept_id)  
8 );
```

emp_id
emp_name
salary
department
join_date
email
phone

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
- information_schema
- mysql
- performance_schema
- phpmyadmin

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0242 seconds.)

```
CREATE TABLE students ( student_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100) NOT NULL, email VARCHAR(100) UNIQUE, age INT, dept_id INT, FOREIGN KEY (dept_id) REFERENCES departments(dept_id) );
```

[Edit inline] [Edit] [Create PHP code]

Console

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.employees:

```
1 CREATE TABLE courses (
2     course_id INT PRIMARY KEY,
3     course_name VARCHAR(100),
4     dept_id INT,
5     FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
6 );
7
```

emp_id
emp_name
salary
department
join_date
email
phone

SELECT* SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query <<

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Console

phpMyAdmin

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - instructors
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: instructors

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0022 seconds.)

`SELECT * FROM `instructors``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

instructor_id	name	email	dept_id
---------------	------	-------	---------

Query results operations

[Create view](#)

Console

phpMyAdmin

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: courses

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.courses:

```
1 CREATE TABLE instructors (  
2   instructor_id INT PRIMARY KEY,  
3   name VARCHAR(100),  
4   email VARCHAR(100) UNIQUE,  
5   dept_id INT,  
6   FOREIGN KEY (dept_id) REFERENCES departments(dept_id)  
7 );
```

course_id
course_name
dept_id

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query <<

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Console

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: instructors

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0022 seconds.)

```
SELECT * FROM `instructors`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

instructor_id	name	email	dept_id
---------------	------	-------	---------

Query results operations

Create view

Console

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: instructors

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.instructors:

```
1 CREATE TABLE enrollments (  
2   student_id INT,  
3   course_id INT,  
4   semester VARCHAR(20),  
5   PRIMARY KEY (student_id, course_id),  
6   FOREIGN KEY (student_id) REFERENCES students(student_id),  
7   FOREIGN KEY (course_id) REFERENCES courses(course_id)  
8 );
```

instructor_id
name
email
dept_id

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query <<

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Console

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: enrollments

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0029 seconds.)

`SELECT * FROM `enrollments``

Query results operations

Create view

Console

2.Insert data and query using SELECT.

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: enrollments

Run SQL query/queries on table college_db.enrollments:

```

1 INSERT INTO departments VALUES (1,'CS'),(2,'EE');
2 INSERT INTO students (name, email, age, dept_id) VALUES
3 ('Ali', 'ali@gmail.com', 20, 1),
4 ('Sara', 'sara@gmail.com', 21, 1),
5 ('Ahmed', 'ahmed@gmail.com', 22, 2);
6 INSERT INTO courses VALUES
7 (101, 'Database', 1),
8 (102, 'AI', 1),
9 (201, 'Circuits', 2);
10 INSERT INTO enrollments VALUES
11 (1, 101, 'Fall 2025'),
12 (1, 102, 'Fall 2025'),
13 (2, 101, 'Fall 2025');
14 SELECT * FROM students;
15 SELECT * FROM courses;

```

student_id
course_id
semester

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Console

phpMyAdmin

Recent Favorites

New

- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - enrollments
 - instructors
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: departments

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)

```
SELECT * FROM `departments`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	dept_id	dept_name
☐ Edit ☐ Copy ☐ Delete	1	CS
☐ Edit ☐ Copy ☐ Delete	2	EE

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

localhost/phpmyadmin/index.php?route=/database/structure&db=phpmyadmin

phpMyAdmin

Recent Favorites

New

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- college_db
 - New
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 - departments
 - employees
 - enrollments
 - instructors
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 2 (3 total, Query took 0.0031 seconds.)

```
SELECT * FROM `students`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	student_id	name	email	age	dept_id
☐ Edit ☐ Copy ☐ Delete	1	Ali	ali@gmail.com	20	1
☐ Edit ☐ Copy ☐ Delete	2	Sara	sara@gmail.com	21	1
☐ Edit ☐ Copy ☐ Delete	3	Ahmed	ahmed@gmail.com	22	2

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

phpMyAdmin

Recent Favorites

New

- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - enrollments
 - instructors
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: courses

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 2 (3 total, Query took 0.0007 seconds.)

```
SELECT * FROM `courses`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

phpMyAdmin

Recent

Favorites

New

collegedb

college_db

New

courses

departments

employees

enrollments

instructors

students

information_schema

mysql

performance_schema

phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: enrollments

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Showing rows 0 - 2 (3 total, Query took 0.0008 seconds.)

SELECT * FROM `enrollments`

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

student_id

course_id

semester

Edit

Copy

Delete

1

101

Fall 2025

Edit

Copy

Delete

1

102

Fall 2025

Edit

Copy

Delete

2

101

Fall 2025

Check all

With selected:

Edit

Copy

Delete

Export

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Console

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Run SQL query/queries on table college_db.students:

1 UPDATE students

student_id

phpMyAdmin

Recent Favorites

New

- collegedb
- college_db
 - New
 - courses
 - departments
 - employees
 - enrollments
 - instructors
 - students
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 2 (3 total, Query took 0.0009 seconds.)

SELECT * FROM `students`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	student_id	name	email	age	dept_id
☐ Edit Copy Delete	1	Ali Khan	ali@gmail.com	20	1
☐ Edit Copy Delete	2	Sara	sara@gmail.com	21	1
☐ Edit Copy Delete	3	Ahmed	ahmed@gmail.com	22	2

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

Console

4.Delete a student record.

phpMyAdmin

Recent Favorites

New

- collegedb

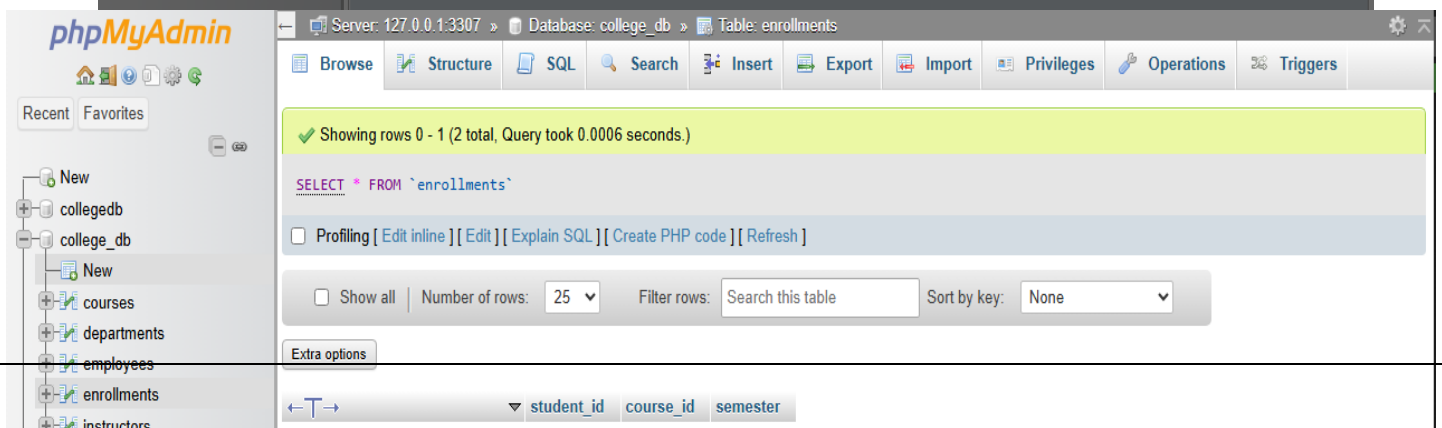
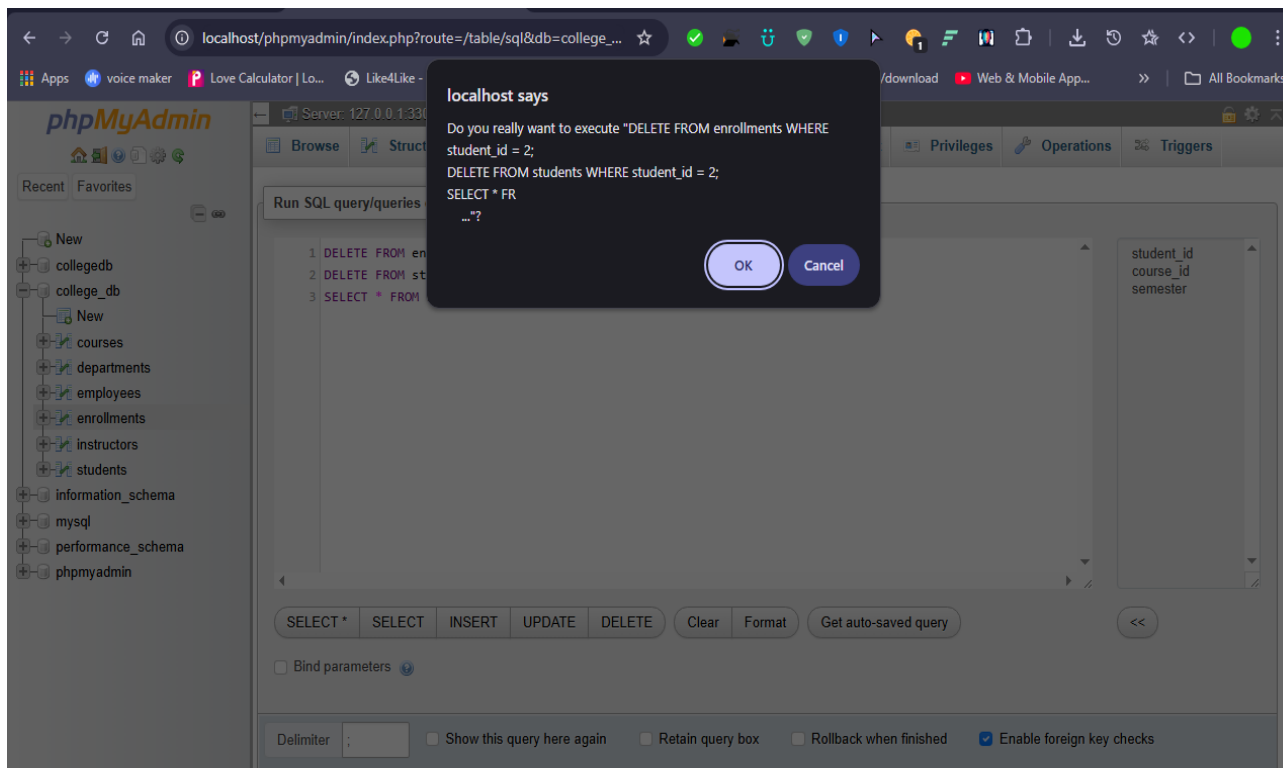
Server: 127.0.0.1:3307 » Database: college_db » Table: enrollments

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.enrollments:

```
1 DELETE FROM enrollments WHERE student_id = 2;
```

student_id



phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 1 (2 total, Query took 0.0010 seconds.)

SELECT * FROM `students`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	student_id	name	email	age	dept_id
☐ Edit Copy Delete	1	Ali Khan	ali@gmail.com	20	1
☐ Edit Copy Delete	3	Ahmed	ahmed@gmail.com	22	2

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Console

5.Try TRUNCATE and DROP.

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.students:

```
1 TRUNCATE TABLE enrollments;
2 TRUNCATE TABLE students;
3
4 DROP TABLE enrollments;
5 DROP TABLE students;
6 DROP TABLE courses;
7 DROP TABLE instructors;
8 DROP TABLE departments;
```

student_id
name
email
age
dept_id

localhost/phpmyadmin/index.php?route=/table/sql&db=college_...

phpMyAdmin

Server: 127.0.0.1:3307

Run SQL query/queries

1 TRUNCATE TABLE enrollments;
2 TRUNCATE TABLE students;
3
4 DROP TABLE enrollments;
5 DROP TABLE students;
6 DROP TABLE courses;
7 DROP TABLE instructors;
8 DROP TABLE departments;

localhost says

Do you really want to execute "TRUNCATE TABLE enrollments;
TRUNCATE TABLE students;

DROP TABLE enrollments;
DROP TABLE students;
D
...";

OK Cancel

student_id
name
email
age
dept_id

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0277 seconds.)

DROP TABLE enrollments;

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0079 seconds.)

-- drop child tables first DROP TABLE students;

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0087 seconds.)

6.Add a new column using ALTER TABLE.

The image displays two screenshots of the phpMyAdmin interface, illustrating the process of adding new columns to a table using the ALTER TABLE command.

Top Screenshot: The interface shows the 'SQL' tab selected for the 'students' table in the 'college_db' database. The SQL query editor contains the following commands:

```
1 ALTER TABLE students
2 ADD phone VARCHAR(20);
3 ALTER TABLE students
4 ADD gpa DECIMAL(3,2) DEFAULT 0.00;
5
6 DESCRIBE students;
```

The right sidebar shows the current structure of the 'students' table with columns: student_id, name, email, age, and dept_id.

Bottom Screenshot: This screenshot shows the execution results of the SQL queries. The first two queries (adding 'phone' and 'gpa' columns) returned an empty result set (zero rows). The final query, 'DESCRIBE students;', shows the updated table structure:

```
DESCRIBE students;
+-----+-----+-----+-----+-----+
| student_id | name | email | age | dept_id |
+-----+-----+-----+-----+-----+
|          |     |      |    |         |
+-----+-----+-----+-----+-----+
```

The interface also displays a confirmation message: "Your SQL query has been executed successfully."

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0014 seconds.)

`SELECT * FROM `students``

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

student_id	name	email	age	dept_id	phone	gpa
------------	------	-------	-----	---------	-------	-----

Query results operations

Create view

Console

7.Practice joins between Students and Courses.

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: students

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.students:

```

1 SELECT
2     s.student_id,
3     s.name      AS student_name,
4     c.course_name,
5     e.semester
6 FROM students s
7 INNER JOIN enrollments e ON s.student_id = e.student_id
8 INNER JOIN courses c     ON e.course_id = c.course_id;
9 SELECT
10    s.name      AS student_name,
11    c.course_name
12 FROM students s

```

student_id	name	email	age	dept_id	phone	gpa
------------	------	-------	-----	---------	-------	-----

8. Explore candidate keys, alternate keys, and composite keys.

Composite Key Example (already in enrollments)
student_id + course_id together = one unique record

```
PRIMARY KEY (student_id, course_id)
```

Candidate Keys in students table:

Both student_id and email can uniquely identify a student

student_id → chosen as PRIMARY KEY

email → becomes ALTERNATE KEY (enforced with UNIQUE)

```
ALTER TABLE students
```

```
ADD CONSTRAINT unique_email UNIQUE (email);
```